EXPERIMENT 4

AIM:
Use function types, scope and closures; apply try-catch; build a palindrome checker

Software/Tools Required:

- Visual Studio Code (VS Code)
- Web Browser (Google Chrome/Microsoft Edge)
- HTML
- JavaScript (ES6)

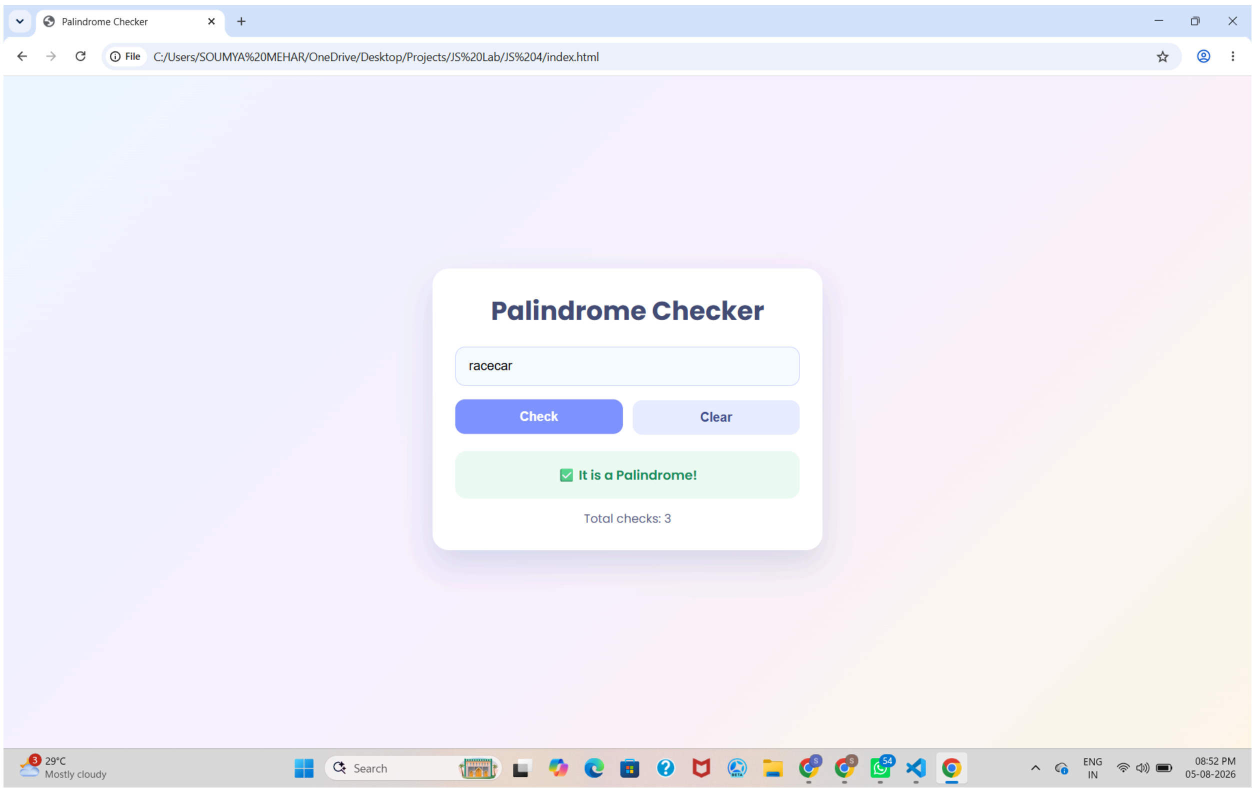
Theory:

A palindrome is a word, number, or sentence that reads the same forward and backward, such as MADAM, LEVEL, and 121. A palindrome checker is a program that checks whether the given input is a palindrome or not by comparing the original string with its reverse.

In JavaScript, different types of functions, such as function declarations, function expressions, and arrow functions, can be used to implement the palindrome checker. Variable scope determines where variables can be accessed in the program, while closures allow an inner function to access variables from its outer function.

The try-catch statement is used for exception handling and helps manage errors, such as invalid or empty input, without stopping the execution of the program. Thus, a palindrome checker demonstrates the concepts of functions, scope, closures, and error handling in JavaScript.

Output:



Result:

Successfully implemented a Palindrome detector using JavaScript ES6 features, including var, let, const, template literals, and destructuring, to find and display whether the given text or number is a palindrome or not.

CASE STUDY 4

AIM:

The ATM system verifies if the pin is palindrome. If the palindrome displays a security message. To implement this create a function to reverse the entered pin number and another function to check if it is palindrome. use function declaration, function expression, arrow function and demonstrate scope and closure concepts.

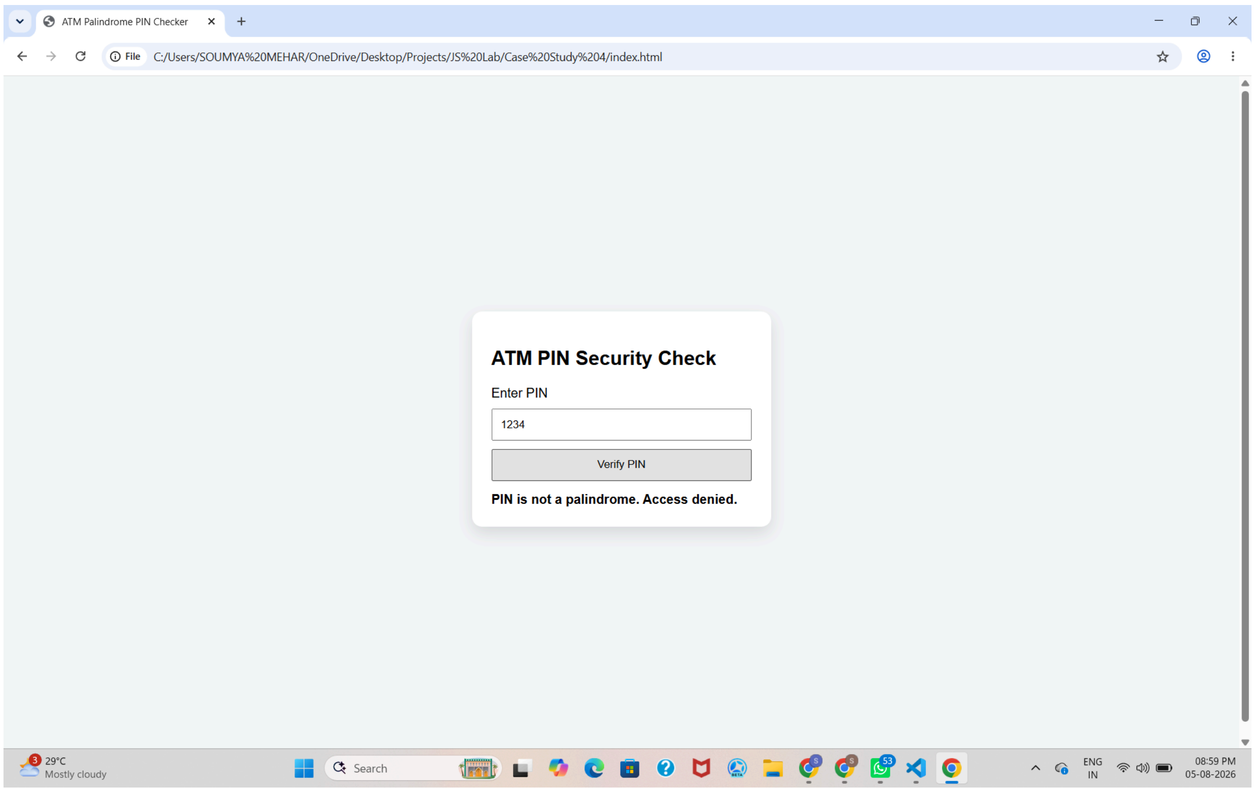
PROGRAM CODE:

```
<!DOCTYPE html>
<html lang="en">
  <head>
    <meta charset="UTF-8" />
    <meta name="viewport" content="width=device-width, initial-scale=1.0" />
    <title>ATM Palindrome PIN Checker</title>
    <style>
      body {
        font-family: Arial, sans-serif;
        background: #f4f6f8;
        display: grid;
        place-items: center;
        min-height: 100vh;
      }
      .card {
        background: white;
        padding: 24px;
        border-radius: 12px;
        box-shadow: 0 8px 20px rgba(0, 0, 0, 0.12);
        width: 320px;
      }
      input,
      button {
        width: 100%;
        padding: 10px;
        margin-top: 10px;
        box-sizing: border-box;
```

```
      }
      #result {
        margin-top: 14px;
        font-weight: bold;
      }
    </style>
  </head>
  <body>
    <div class="card">
      <h2>ATM PIN Security Check</h2>
      <label for="pinInput">Enter PIN</label>
      <input id="pinInput" type="text" placeholder="e.g. 121" />
      <button id="checkBtn">Verify PIN</button>
      <div id="result">Result will appear here</div>
    </div>

    <script src="script.js"></script>
  </body>
</html>
```

OUTPUT:



RESULT:

Successfully implemented a ATM pin Palindrome detector using JavaScript ES6 features, including var, let, const, template literals, and destructuring, to find and display whether the given ATM pin is a palindrome or not.